

Browser Extension Email Testing

1. Artefact Purpose

This technical artefact documents browser-extension testing of the implemented phishing-email detection system. The supplied evidence contains twelve controlled tests captured from the testing interface.

The artefact demonstrates integration and end-to-end operation rather than only model performance. The browser extension extracts the currently displayed controlled email, sends the relevant content to the local inference service and presents the returned classification and contextual information beside the email.

2. Controlled Test Environment

The screenshots show the browser running the controlled dataset page at localhost.

Component	Role
Dataset page interface	Provides a repeatable mail-client-style environment containing selected held-out emails.
Browser extension popup	Provides the Analyse Current Email action and displays the prediction beside the currently viewed message.
Content extraction	Reads the controlled page's sender, subject and body fields.
FastAPI backend	Receives the request and runs the deployed classifier.
DeBERTa-v3-large gated-hybrid model	Produces Safe/Phishing class probabilities using contextual text representation plus engineered phishing features.
Rule/context layer	Displays supplementary Email Characteristics and Phishing Indicators.
Detection Method panel	Explains that contextual/semantic model analysis is supplemented by rule-based indicators.

3. End-to-End Browser Extension Methodology

- 1) A held-out email is selected in the controlled inbox.
- 2) The extension identifies the currently visible sender, subject and body from the controlled page.
- 3) The user selects Analyse Current Email in the extension popup.
- 4) The extension sends the extracted email content to the local FastAPI service.

- 5) The backend constructs the model text from the subject and body and performs DeBERTa-v3-large gated-hybrid inference.
- 6) The API returns the predicted class and Safe/Phishing probabilities.
- 7) The extension renders the verdict, confidence, sender/subject, neutral email characteristics, phishing indicators and detection-method explanation.
- 8) The displayed ground truth remains part of the controlled test page and is used for evaluation comparison; it is not treated as model input.

3.1 What the Extension Does:

The extension is the browser-facing integration layer. It does not independently classify the email. Classification occurs in the backend model. The extension's responsibility is extraction, API communication and presentation of the result. This separation is important because it allows the same inference service to support both the Streamlit interface and browser-based testing.

3.2 Understanding the Popup

Popup element	Meaning
Analyse Current Email	Triggers extraction and inference for the email currently open in the controlled reader.
Safe Email / Phishing Email	The model's final displayed class.
Confidence	Probability assigned to the selected class for that email.
Safe / Phishing percentages	Two-class probability breakdown returned by the model.
Email Details	Confirms the sender and subject associated with the analysed view.
Email Characteristics	Neutral structural observations such as email-address, URL or phone presence.
Phishing Indicators	Supplementary contextual/rule-based warnings; these are not the classifier itself.
Detection Method	Explains that DeBERTa-v3-large performs contextual/semantic analysis and rule-based indicators provide supporting context.

4. Evidence Summary

Case	Screenshot test	Dataset item	Displayed subject	Controlled ground truth
BE-01	Test 1	Email 31	Security Notification 31	Phishing Email
BE-02	Test 2	Email 32	University Services Update 32	Safe Email
BE-03	Test 3	Email 33	Security Notification 33	Phishing Email
BE-04	Test 4	Email 34	University Services Update 34	Safe Email
BE-05	Test 5	Email 35	Security Notification 35	Phishing Email
BE-06	Test 6	Email 36	University Services Update 36	Safe Email
BE-07	Test 7	Email 37	Security Notification 37	Phishing Email
BE-08	Test 8	Email 38	University Services Update 38	Safe Email
BE-09	Test 9	Email 39	Security Notification 39	Phishing Email
BE-10	Test 10	Email 40	University Services Update 40	Safe Email
BE-11	Test 11	Email 41	Security Notification 41	Phishing Email
BE-12	Test 12	Email 42	University Services Update 42	Safe Email

The supplied evidence alternates phishing and safe held-out examples from Email 31 through Email 40, followed by Email 49 and Email 50. The screenshots therefore demonstrate both output states repeatedly in the same extension workflow.

5. Browser Extension Tests

5.1 Browser Extension Test 1

Controlled expected class: Phishing Email. The purpose of this test is to verify the complete browser path from selection of Email 31, through extension extraction and backend inference, to presentation of the verdict and contextual information.

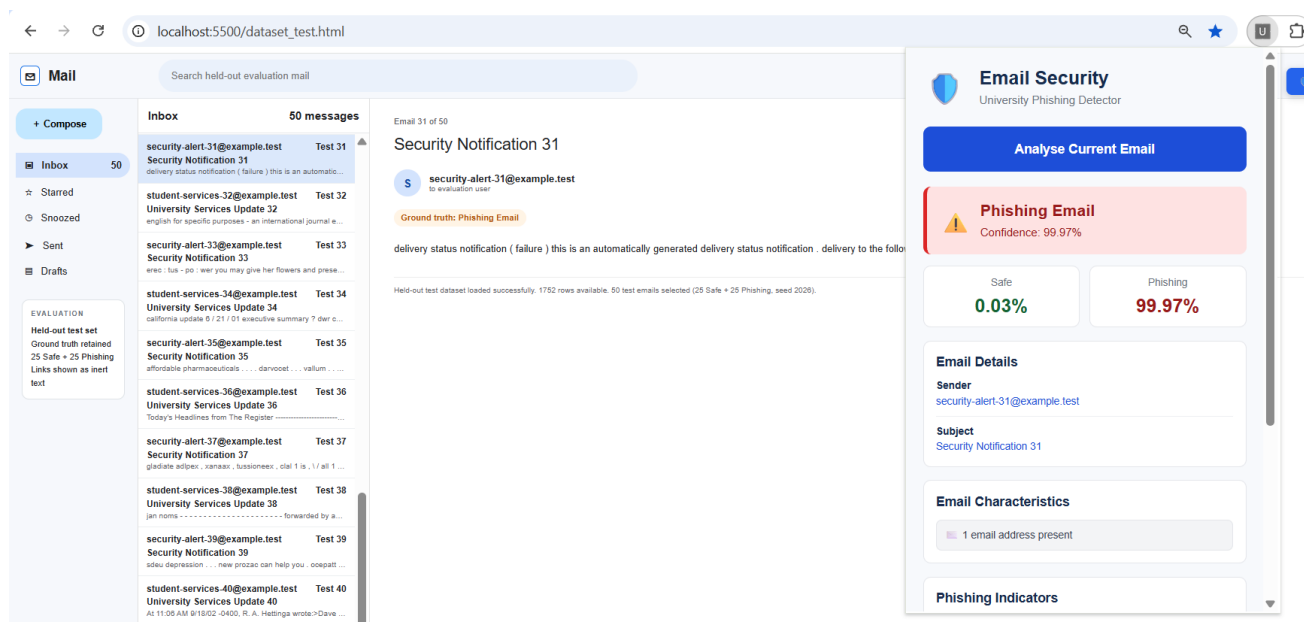


Figure 5.1. Test 1 - primary extension result.

Test 1 opens Email 31, Security Notification 31. The controlled reader visibly marks the ground truth as Phishing Email. The extension returns Phishing Email with a very high phishing probability and reproduces the sender/subject in its Email Details section.

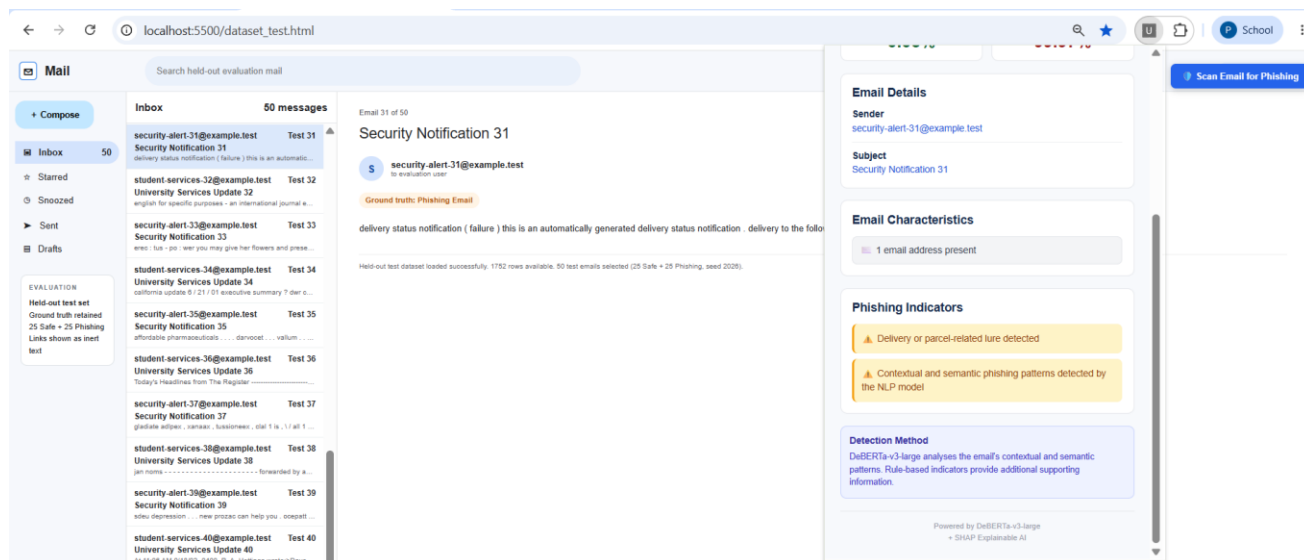


Figure 5.2 Test 1 - characteristics, indicators and detection method.

The scrolled popup shows one email address as a structural characteristic. It also reports a delivery/parcel-related lure and contextual/semantic phishing patterns. The Detection Method panel clarifies that the contextual DeBERTa analysis is supplemented by rule-based information.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

5.2 Browser Extension Test 2

Controlled expected class: Safe Email. The purpose of this test is to verify the complete browser path from selection of Email 32, through extension extraction and backend inference, to presentation of the verdict and contextual information.

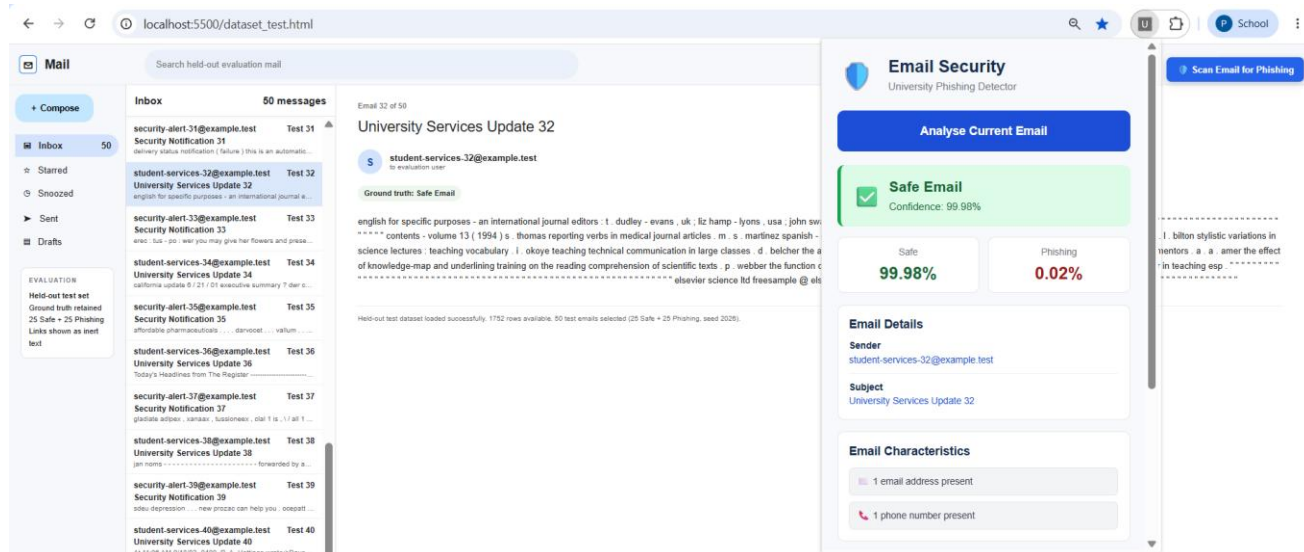


Figure 5.3 Test 2 - primary extension result.

Test 2 opens Email 32, University Services Update 32. The extension returns Safe Email with a very high Safe probability. The same sender and subject are shown in the popup, demonstrating that the extension analysed the currently selected message rather than another inbox item.

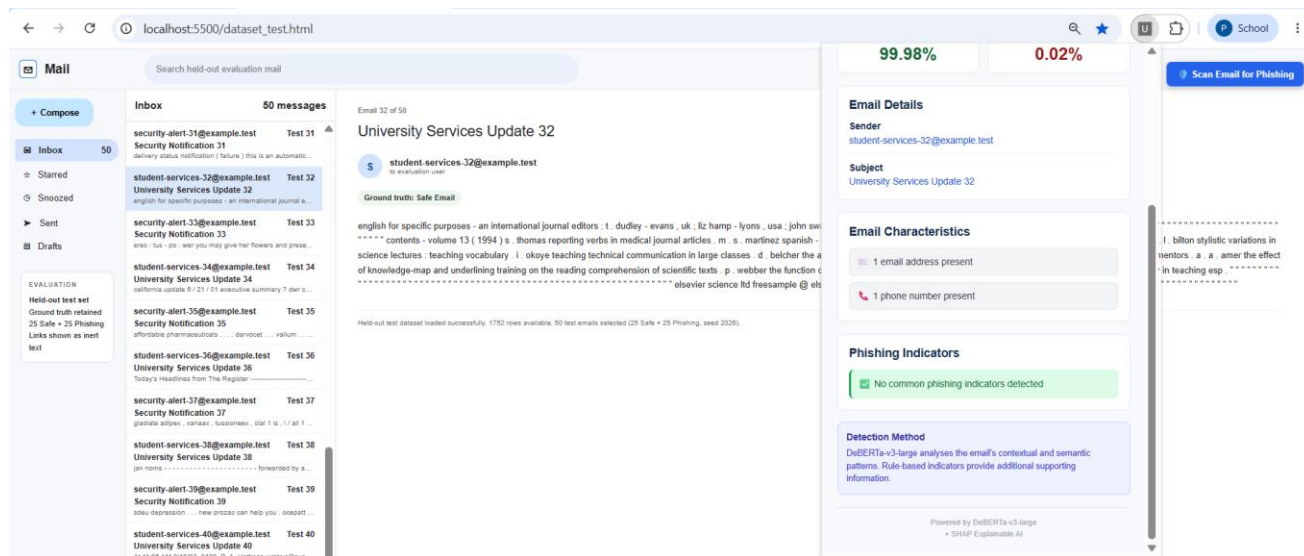


Figure 5.4 Test 2 - characteristics, indicators and detection method.

The lower popup view shows neutral characteristics and a green 'no common phishing indicators' state. This is useful evidence that the interface can return a clean contextual assessment rather than always producing warning banners.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

5.3 Browser Extension Test 3

Controlled expected class: Phishing Email. The purpose of this test is to verify the complete browser path from selection of Email 33, through extension extraction and backend inference, to presentation of the verdict and contextual information.

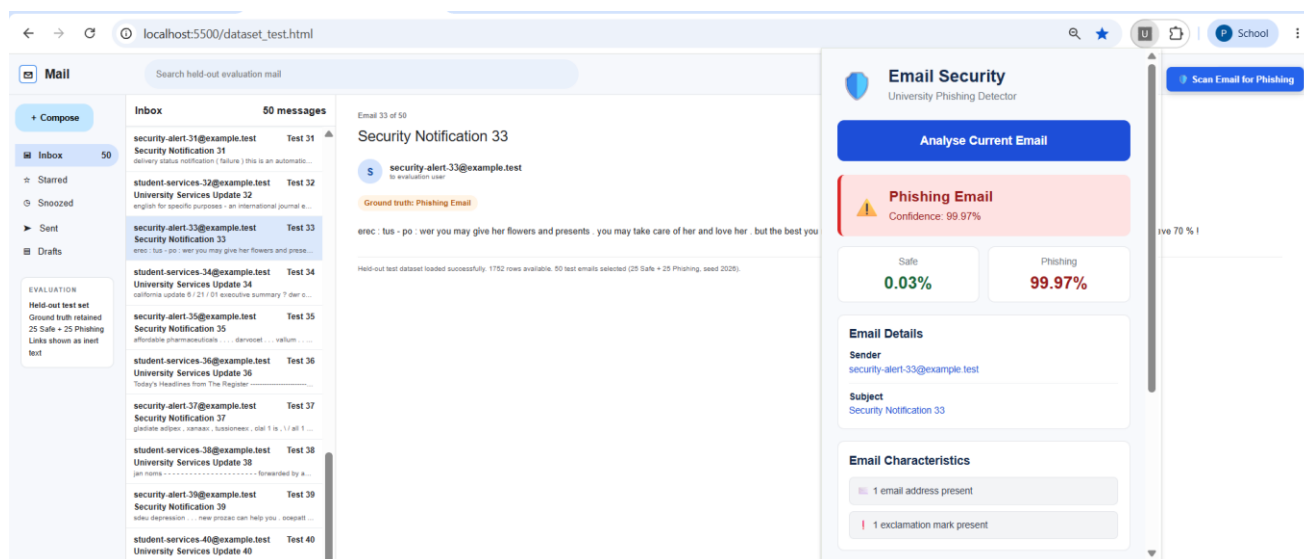


Figure 5.5 Test 3 - primary extension result.

Test 3 analyses Email 33, Security Notification 33. The controlled item is a phishing-labelled example and the extension displays Phishing Email with a high phishing probability.

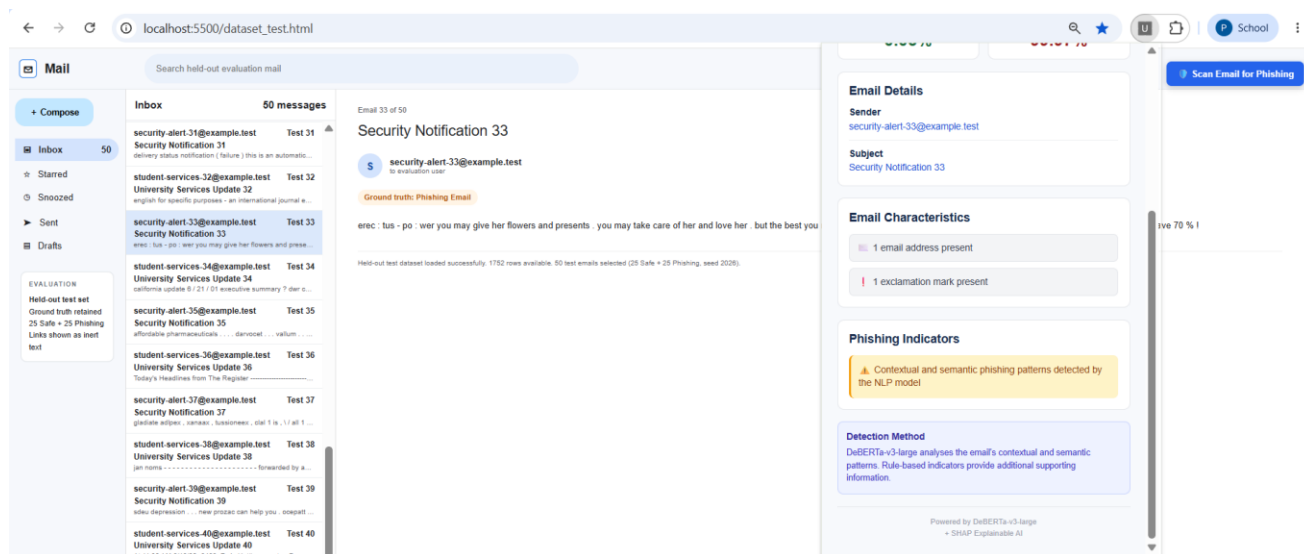


Figure 5.6 Test 3 - characteristics, indicators, and detection method.

The lower view exposes the contextual warning area and Detection Method panel. The evidence shows that the browser integration presents both the model verdict and supporting contextual information without treating the indicator layer as a separate final classifier.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

5.4 Browser Extension Test 4

Controlled expected class: Safe Email. The purpose of this test is to verify the complete browser path from selection of Email 34, through extension extraction and backend inference, to presentation of the verdict and contextual information.

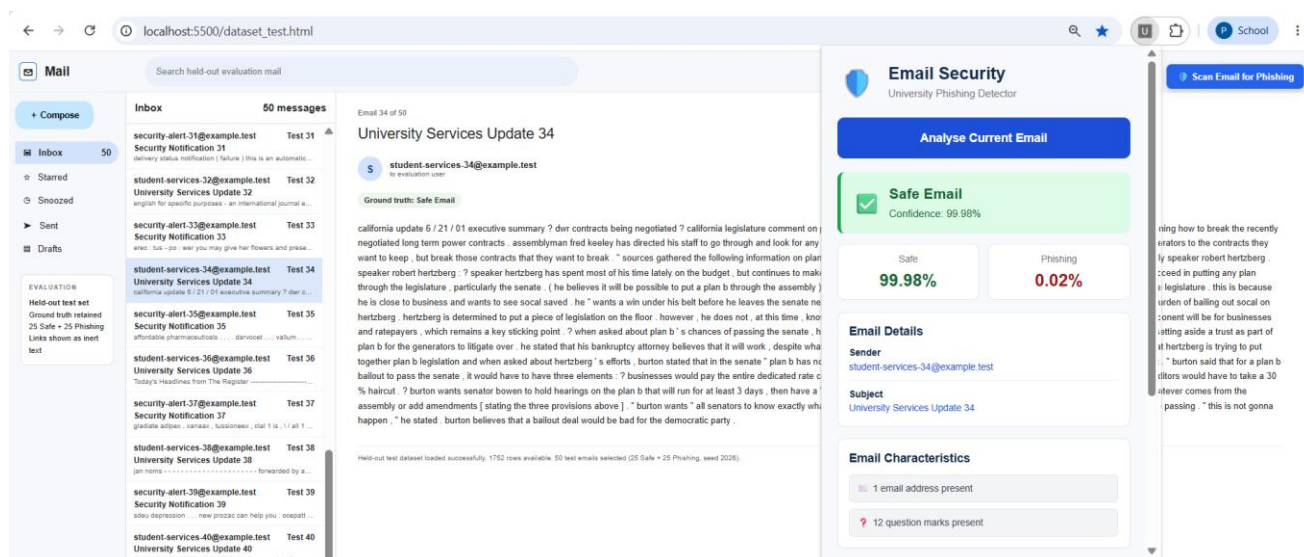


Figure 5.7 Test 4 - primary extension result.

Test 4 analyses Email 34, University Services Update 34, a substantially longer safe-style message. The extension returns Safe Email with a very high Safe probability.

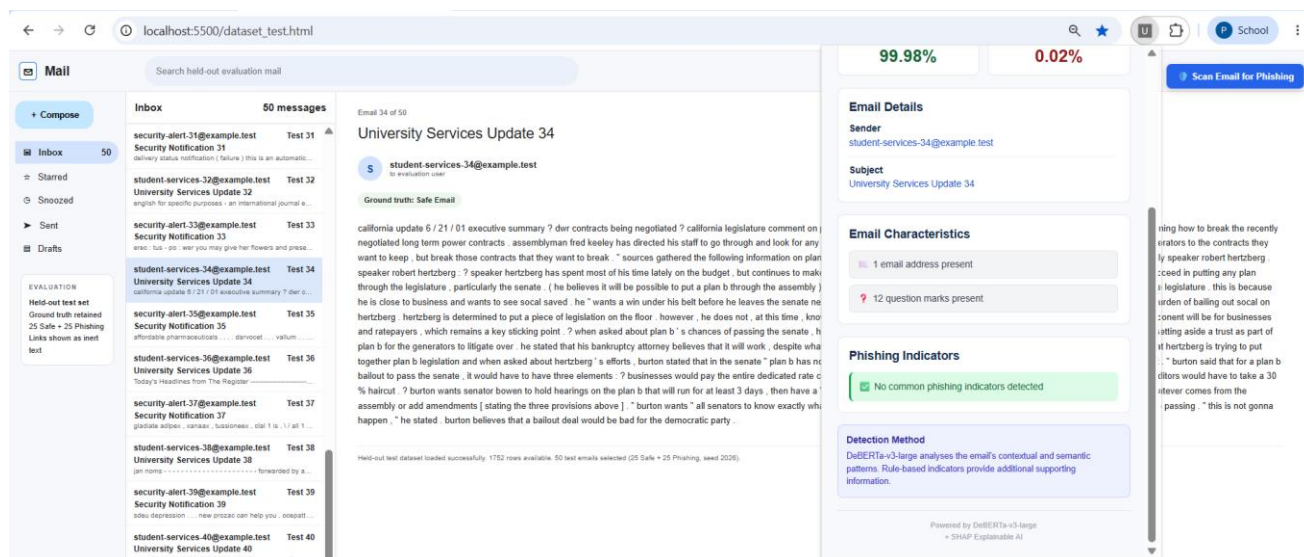


Figure 5.8 Test 4 - characteristics, indicators and detection method.

The second view shows a green no-common-phishing-indicators result and the detection-method explanation. This demonstrates that long-form text can be processed through the same browser workflow and that the popup remains usable through scrolling.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

5.5 Browser Extension Test 5

Controlled expected class: Phishing Email. The purpose of this test is to verify the complete browser path from selection of Email 35, through extension extraction and backend inference, to presentation of the verdict and contextual information.

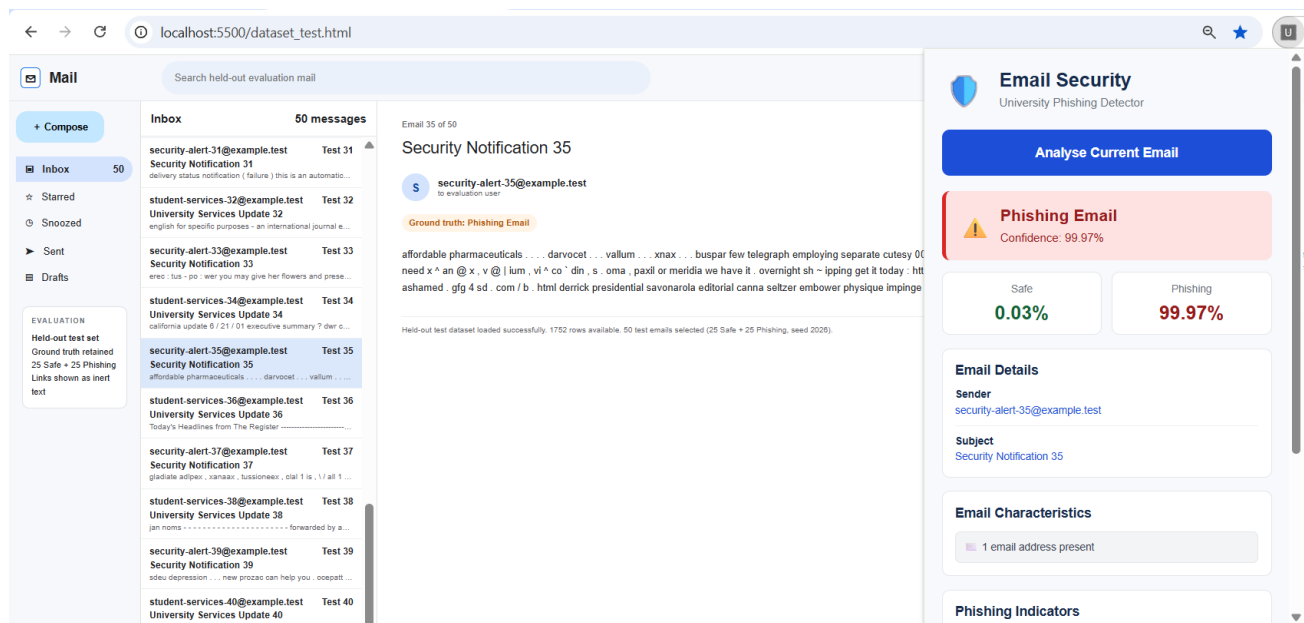


Figure BE5.9. Test 5 - primary extension result.

Test 5 analyses Email 35, Security Notification 35. The extension returns Phishing Email with a high phishing probability while preserving the analysed sender and subject in the popup.

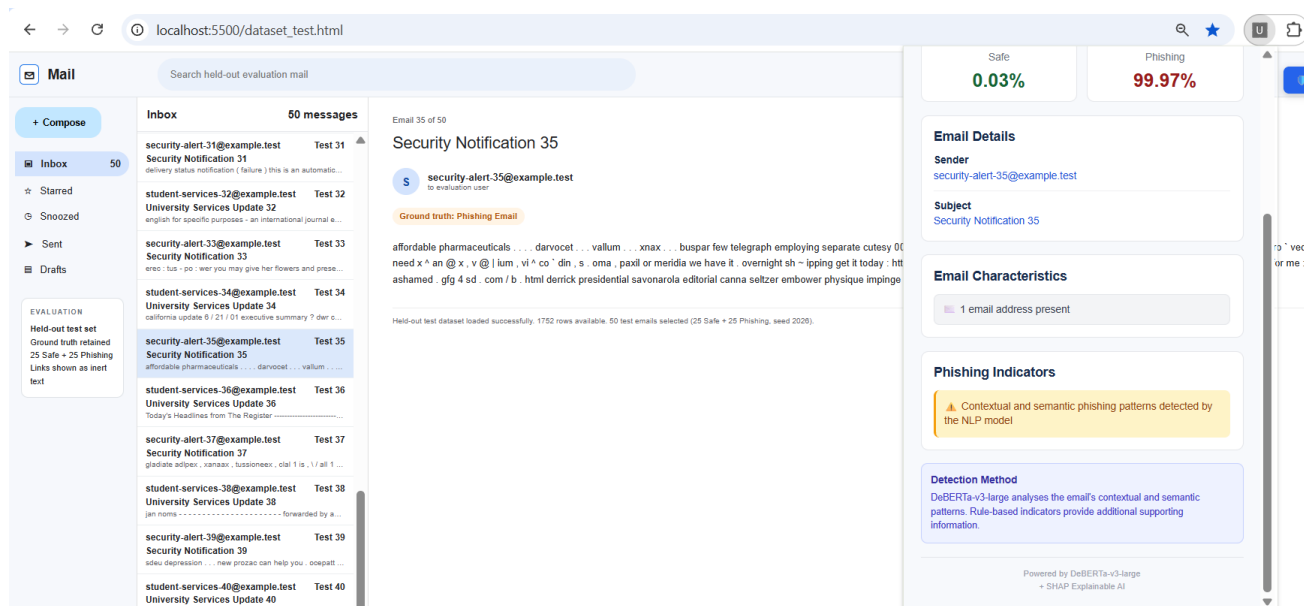


Figure 5.10 Test 5 - characteristics, indicators and detection method.

The lower section reports a contextual/semantic phishing warning. This demonstrates a model-led detection path where the learned representation supplies the main verdict and the contextual panel communicates supporting information.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

5.6 Browser Extension Test 6

Controlled expected class: Safe Email. The purpose of this test is to verify the complete browser path from selection of Email 36, through extension extraction and backend inference, to presentation of the verdict and contextual information.

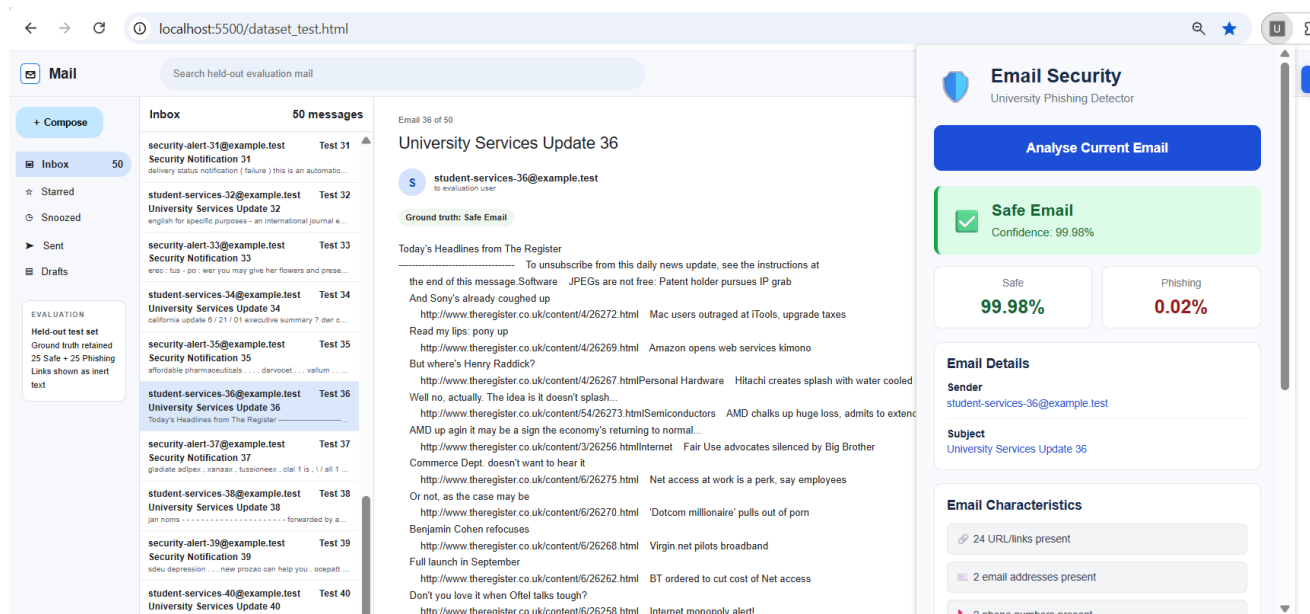


Figure 5.11. Test 6 - primary extension result.

Test 6 analyses Email 36, University Services Update 36. The extension returns Safe Email with a high Safe probability. The email body contains a longer register/list-style message, broadening the qualitative test beyond short prose.

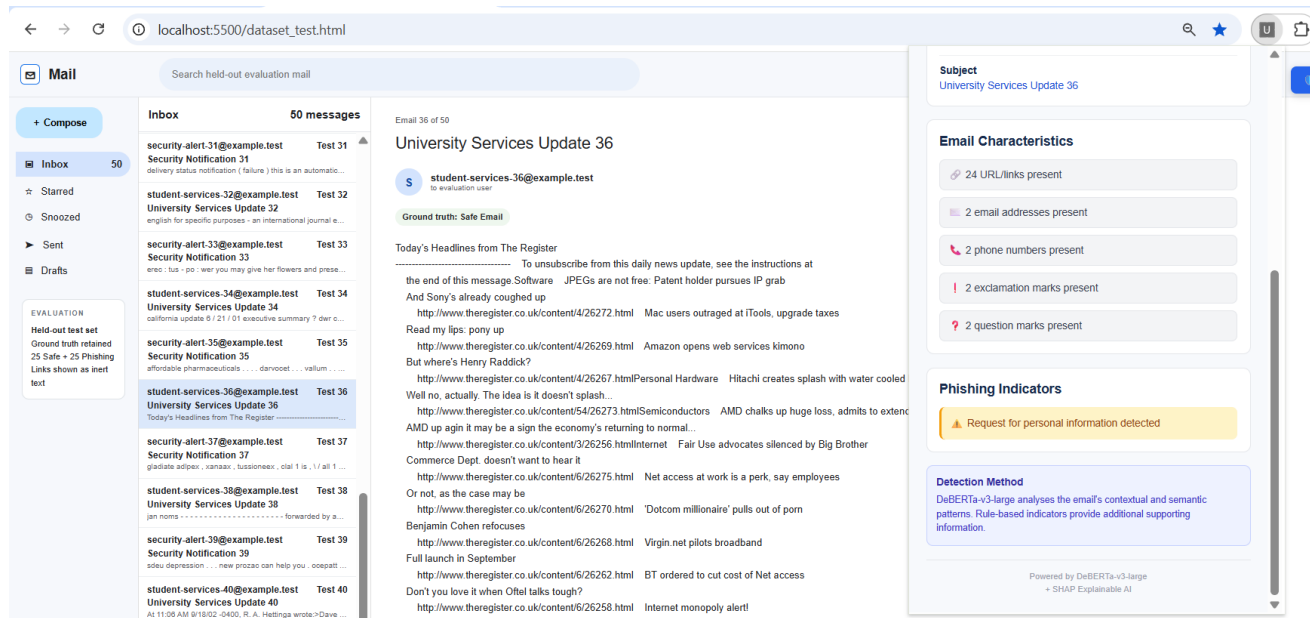


Figure 5.12. Test 6 - characteristics, indicators and detection method.

The scrolled popup exposes multiple neutral characteristics and a separate warning/indicator area. This screenshot is useful because it demonstrates that characteristics such as URLs,

email addresses or phone-like patterns are contextual features and do not automatically force a phishing verdict.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

5.7 Browser Extension Test 7

Controlled expected class: Phishing Email. The purpose of this test is to verify the complete browser path from selection of Email 37, through extension extraction and backend inference, to presentation of the verdict and contextual information.

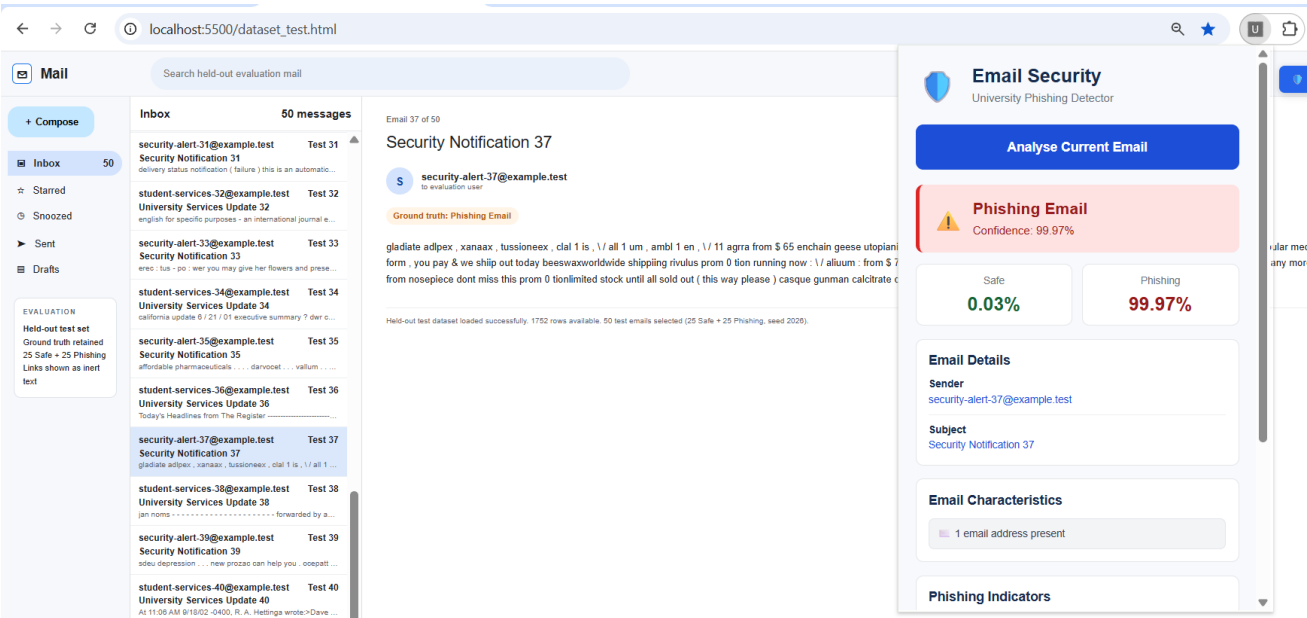


Figure 5.13. Test 7 - primary extension result.

Test 7 analyses Email 37, Security Notification 37. The extension returns Phishing Email with a high phishing probability.

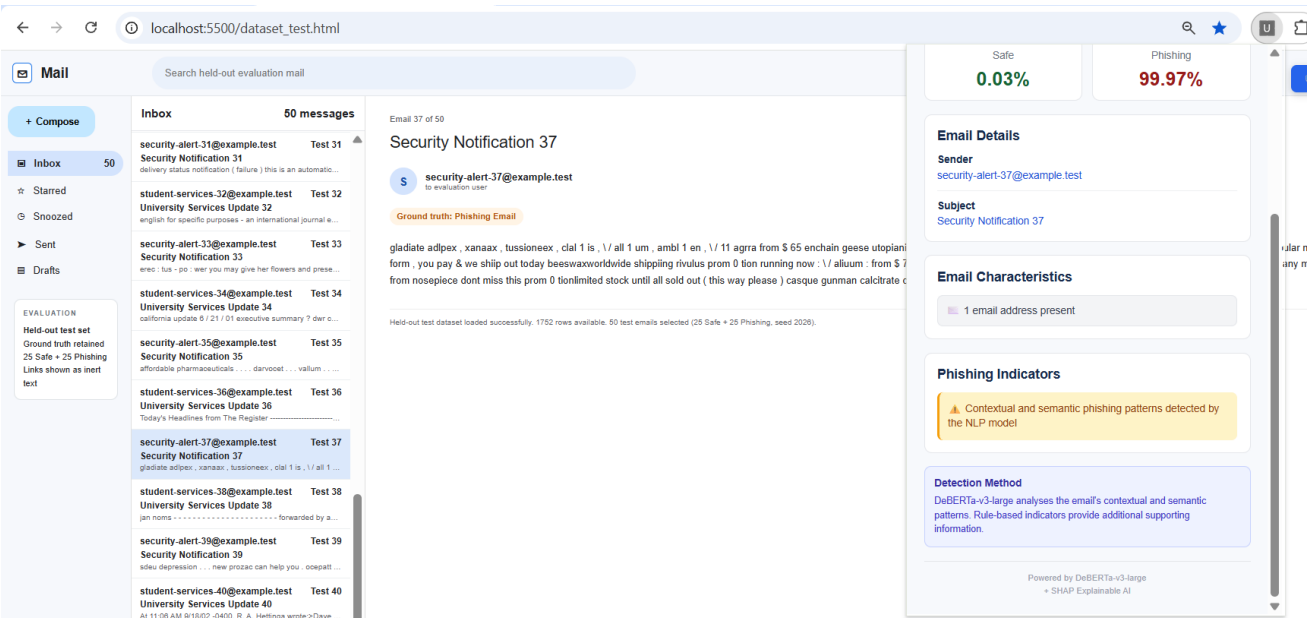


Figure 5.14. Test 7 - characteristics, indicators and detection method.

The lower view reports contextual/semantic phishing patterns and the Detection Method panel. The screenshot demonstrates the same end-to-end flow on another phishing-labelled message, supporting repeatability of extraction, API inference and rendering.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

5.8 Browser Extension Test 8

Controlled expected class: Safe Email. The purpose of this test is to verify the complete browser path from selection of Email 38, through extension extraction and backend inference, to presentation of the verdict and contextual information.

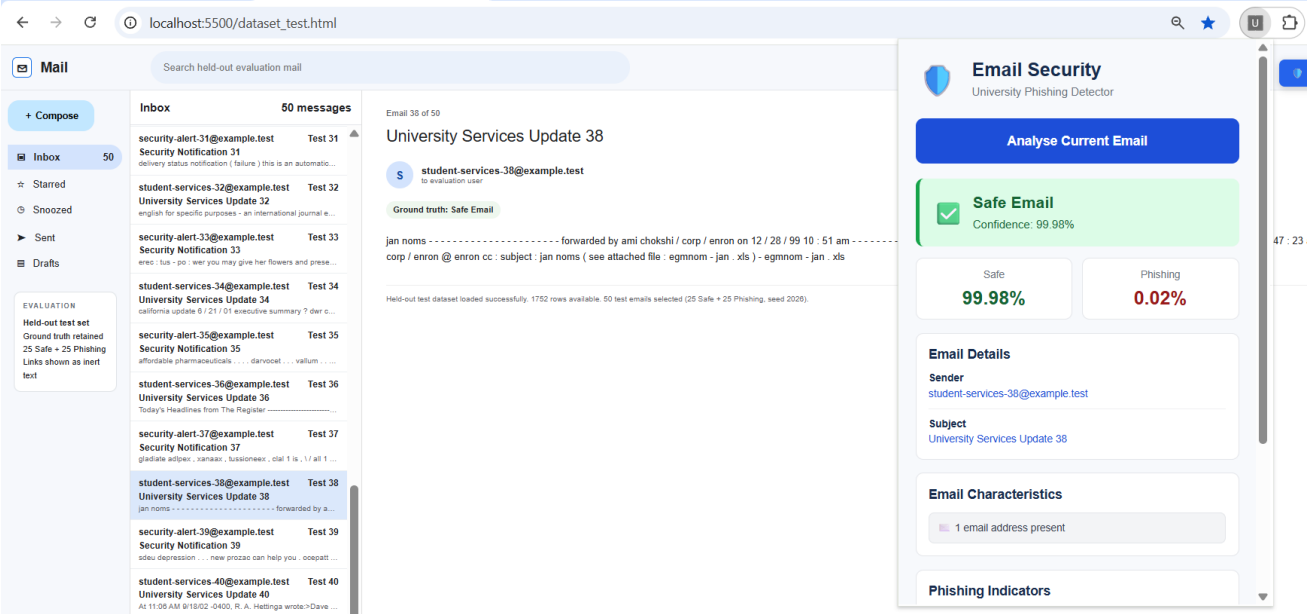


Figure 5.15. Test 8 - primary extension result.

Test 8 analyses Email 38, University Services Update 38. The extension returns Safe Email with a high Safe probability.

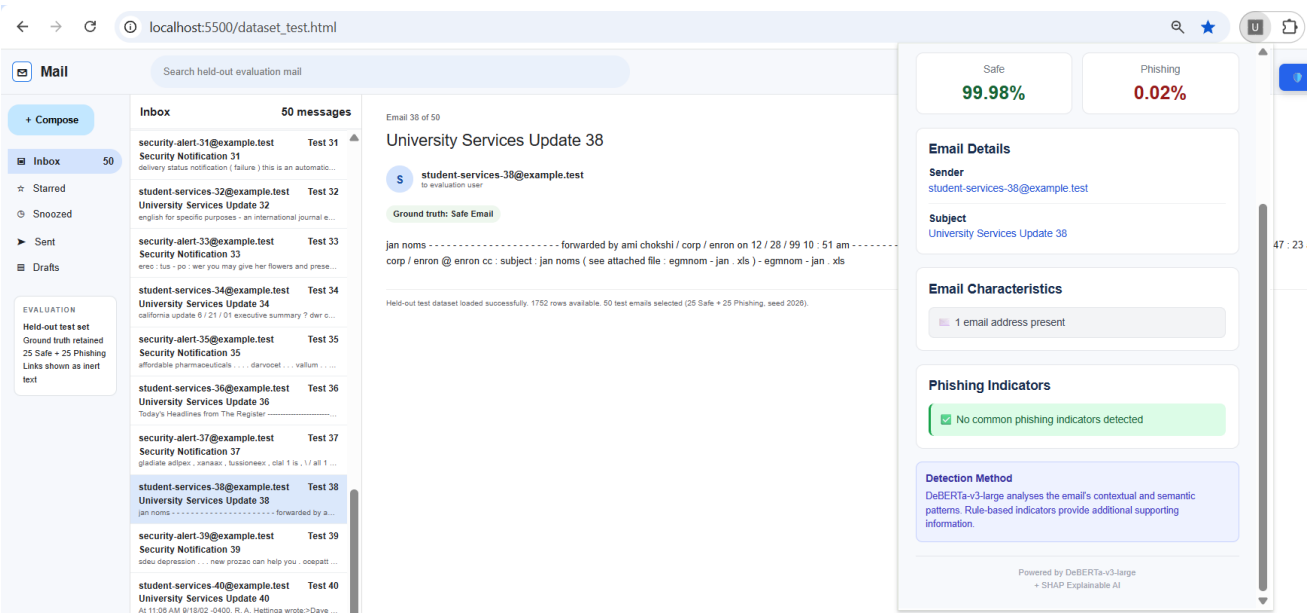


Figure 5.16. Test 8 - characteristics, indicators and detection method.

The lower popup view reports no common phishing indicators. This provides a direct safe counterpart to the preceding phishing case and demonstrates that the same browser extension can represent both outcomes consistently.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

5.9 Browser Extension Test 9

Controlled expected class: Phishing Email. The purpose of this test is to verify the complete browser path from selection of Email 39, through extension extraction and backend inference, to presentation of the verdict and contextual information.

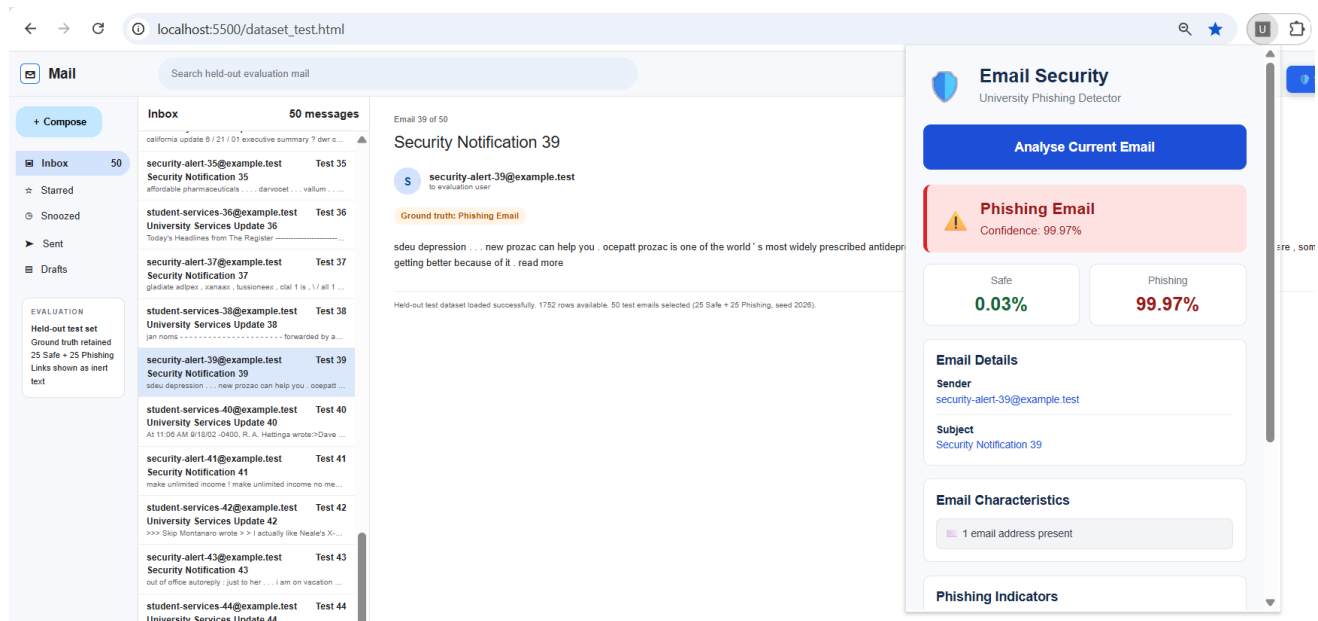


Figure 5.17. Test 9 - primary extension result.

Test 9 analyses Email 39, Security Notification 39. The extension returns Phishing Email with a high phishing probability.

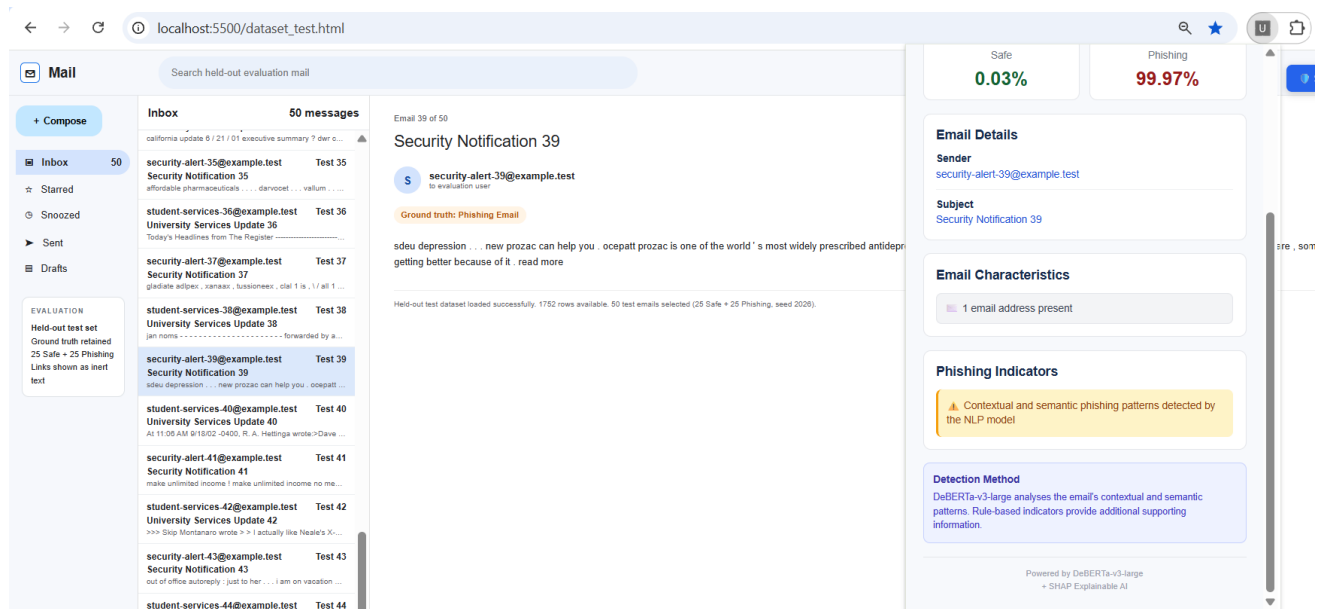


Figure 5.18. Test 9 - characteristics, indicators and detection method.

The lower view shows the phishing-indicator area and contextual model explanation. The evidence demonstrates that the verdict is not merely a visual label on the test page: the

extension independently presents its returned probability breakdown and model-oriented detection information.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

5.10 Browser Extension Test 10

Controlled expected class: Safe Email. The purpose of this test is to verify the complete browser path from selection of Email 40, through extension extraction and backend inference, to presentation of the verdict and contextual information.

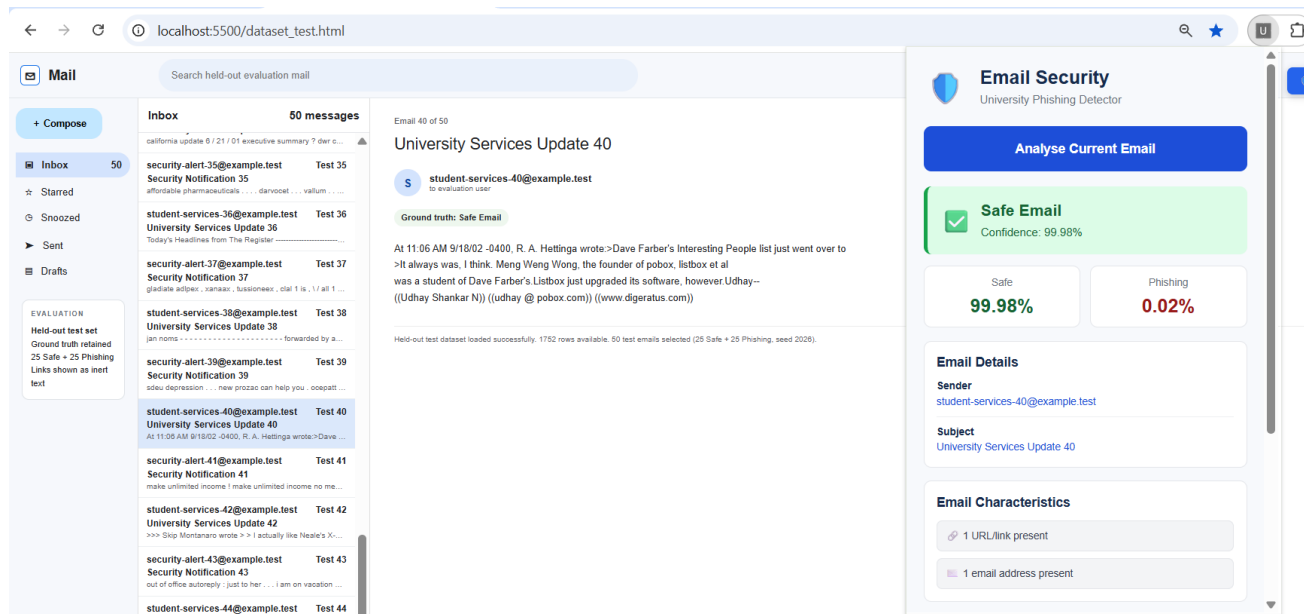


Figure 5.19. Test 10 - primary extension result.

Test 10 analyses Email 40, University Services Update 40. The extension returns Safe Email with a high Safe probability and confirms the associated sender and subject.

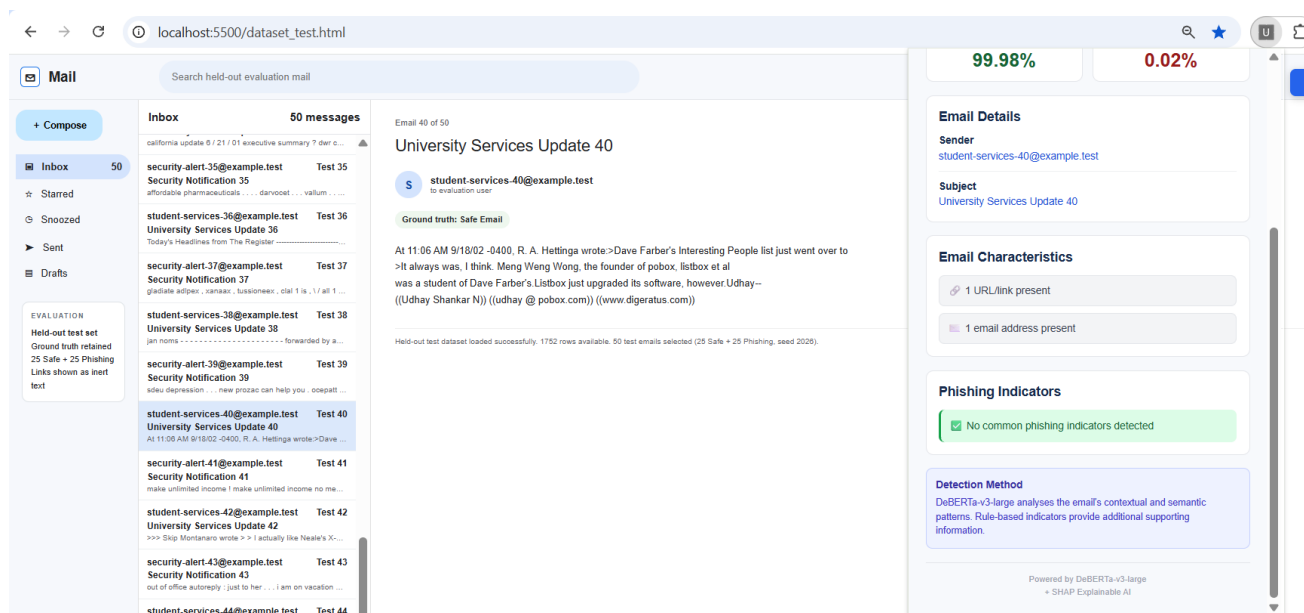


Figure 5.20. Test 10 - characteristics, indicators and detection method.

The second view shows a green no-common-phishing-indicators state and the Detection Method panel. The result provides another repeatable safe classification through the controlled browser workflow.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

5.11 Browser Extension Test 11

Controlled expected class: Phishing Email. The purpose of this test is to verify the complete browser path from selection of Email 49, through extension extraction and backend inference, to presentation of the verdict and contextual information.

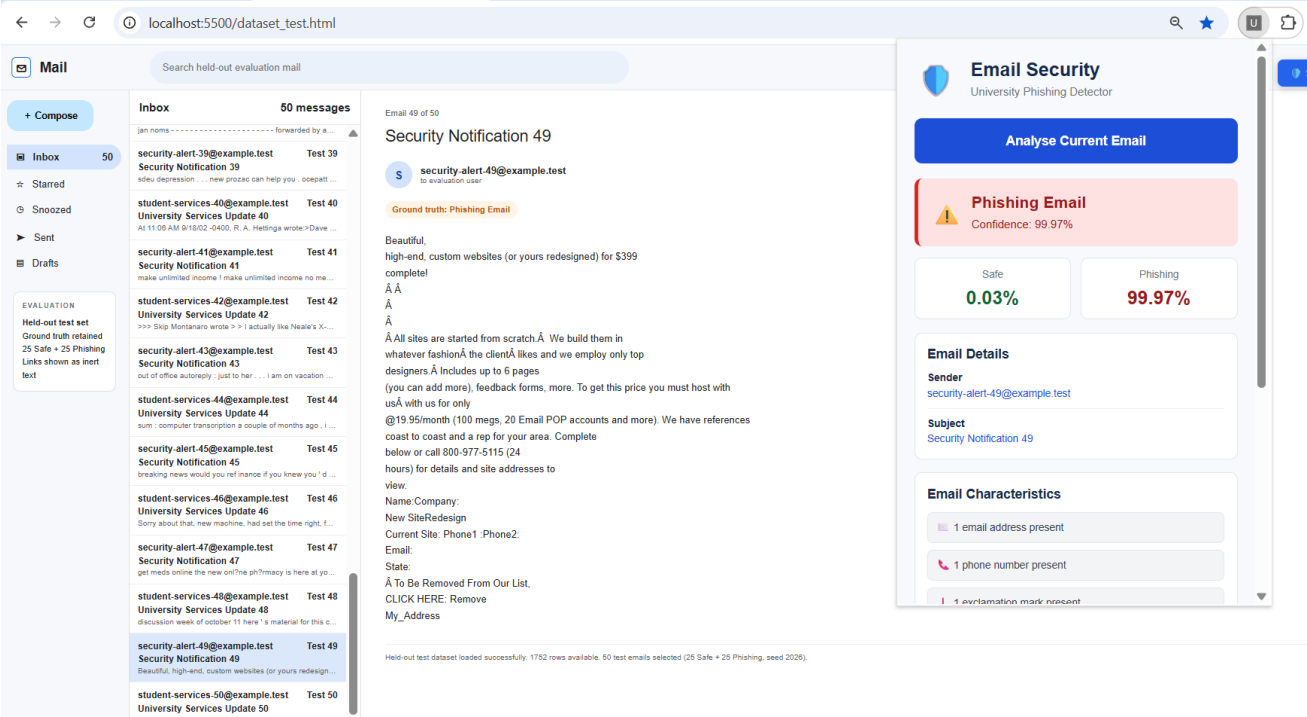


Figure 5.21. Test 11 - primary extension result.

Test 11 jumps to Email 49, Security Notification 49. The extension returns Phishing Email with a very high phishing probability. The message body is visibly different in structure from earlier examples, providing a broader qualitative test.

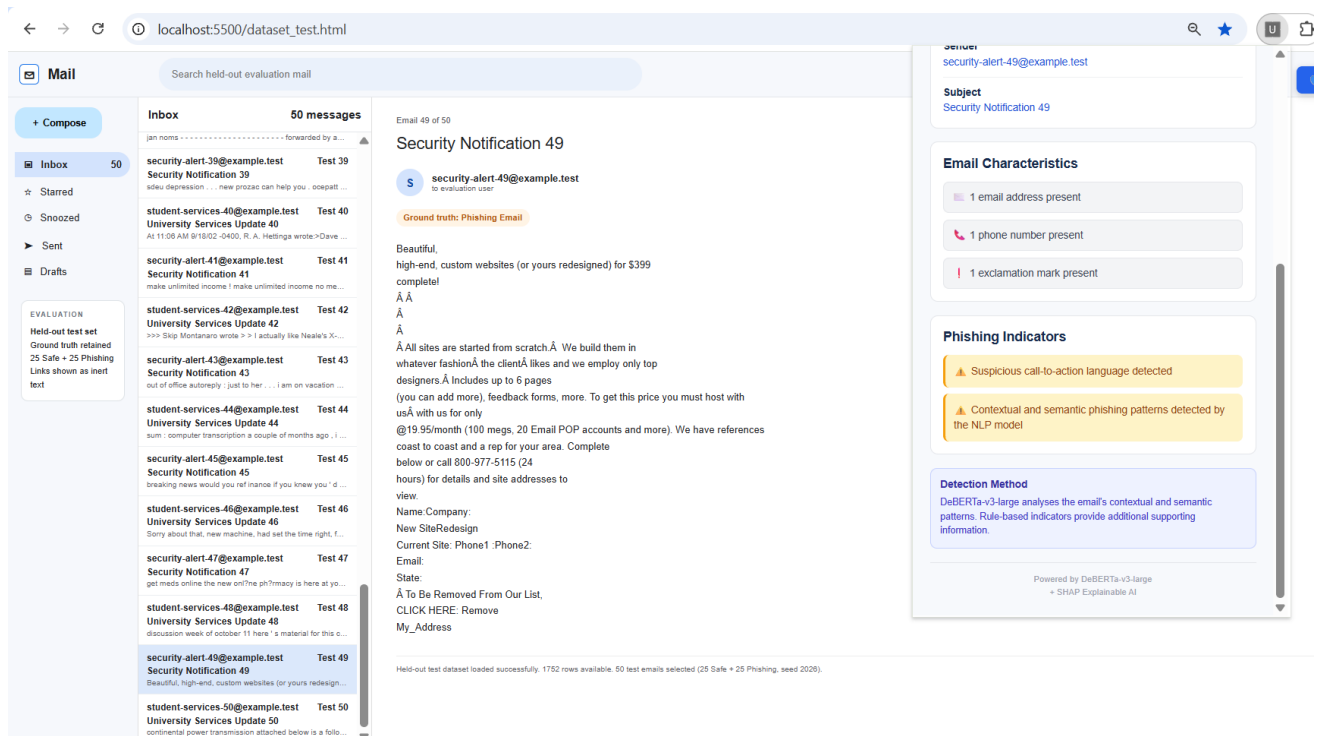


Figure BE22. Test 11 - characteristics, indicators and detection method.

The lower view shows multiple email characteristics and more than one phishing indicator, including contextual/semantic model evidence. This demonstrates that the popup can surface several contextual findings while keeping the final model verdict visually distinct.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

5.12 Browser Extension Test 12

Controlled expected class: Safe Email. The purpose of this test is to verify the complete browser path from selection of Email 50, through extension extraction and backend inference, to presentation of the verdict and contextual information.

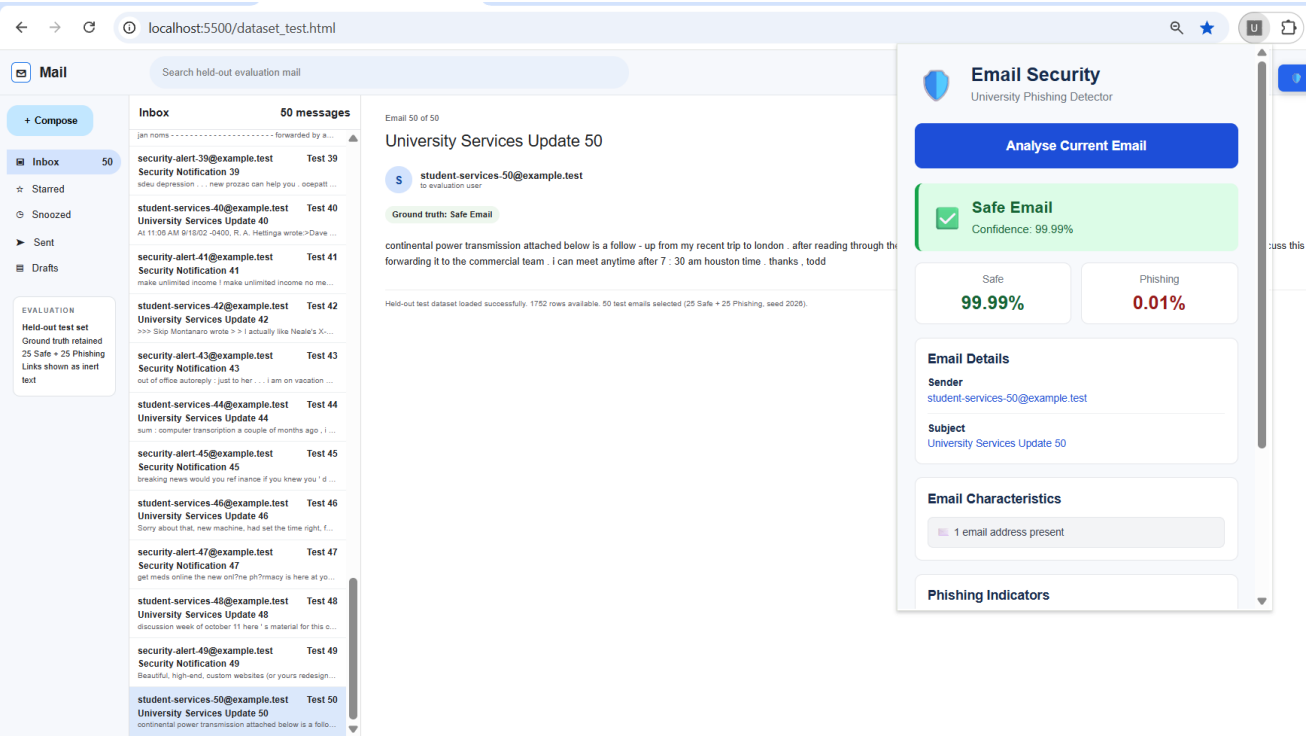


Figure 5.23. Test 12 - primary extension result.

Test 12 analyses Email 50, University Services Update 50. The extension returns Safe Email with a very high Safe probability.

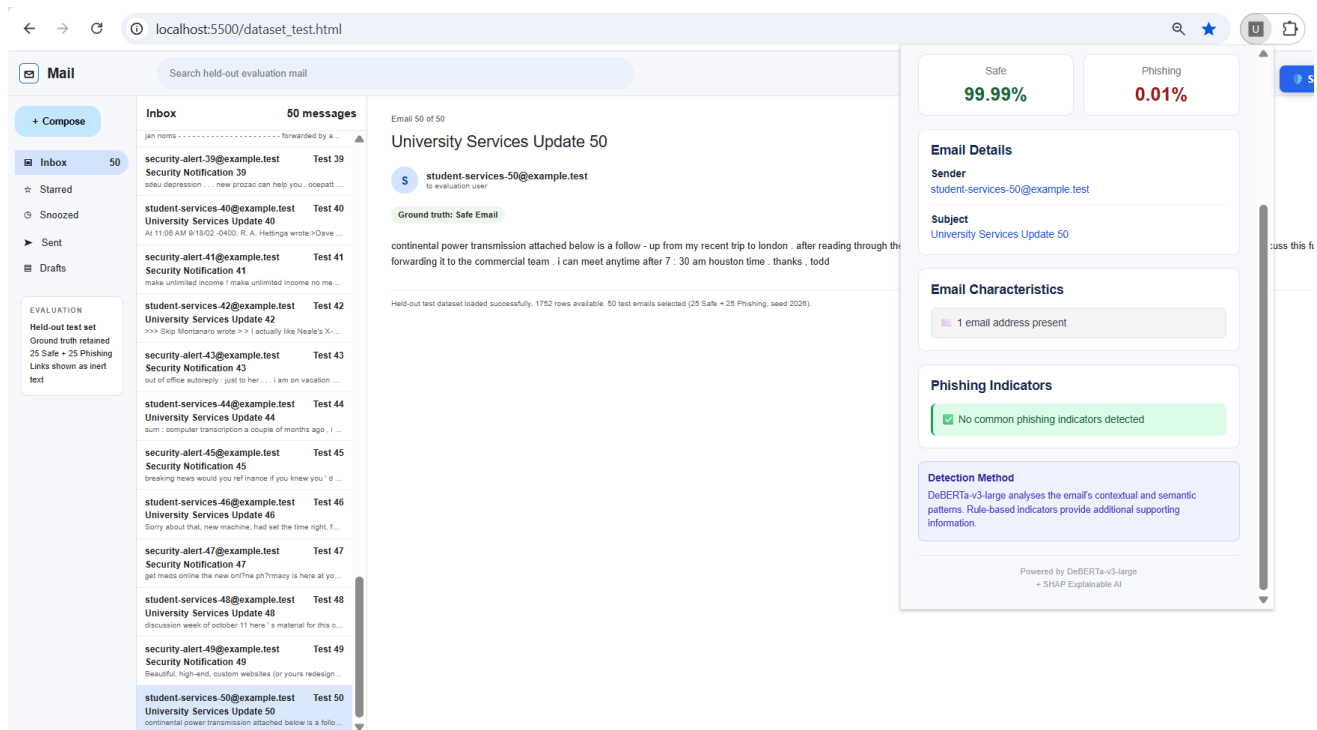


Figure 5.24. Test 12 - characteristics, indicators and detection method.

The lower view reports no common phishing indicators and displays the detection-method explanation. This final case completes the alternating demonstration with a safe result and confirms that the extension continues to extract and render the correct currently selected message.

Test interpretation

Functional outcome: the extension is visibly associated with the currently open controlled email and presents the model output in the browser without requiring the user to copy the message into the Streamlit interface. The evidence should be interpreted as controlled end-to-end browser integration testing.

6. Cross-Test Technical Findings

- The extension successfully demonstrates both Safe and Phishing result states across the supplied controlled examples.
- The popup repeats sender and subject information, providing a visible check that the currently open email was the item analysed.
- The probability breakdown gives more information than a binary label and allows confidence to be inspected.
- Email Characteristics are deliberately neutral observations; their presence does not independently mean phishing.
- Phishing Indicators are supplementary contextual warnings and remain visually separate from the model's final verdict.
- The Detection Method panel communicates the hybrid design: contextual/semantic DeBERTa analysis with rule-based indicators as supporting information.
- The scrollable popup allows the interface to present prediction, metadata, characteristics and indicators without obscuring the underlying email.
- The controlled page retains ground truth for evaluator comparison, enabling the browser integration to be demonstrated against known held-out examples.

7. Functional Acceptance Assessment

Requirement	Evidence	Assessment
Extract the currently displayed controlled email	Each test shows the selected inbox item and matching sender/subject in the extension.	Pass
Invoke backend analysis from the browser	Analyse Current Email produces a populated result card.	Pass
Display both model classes	The evidence contains repeated Safe and Phishing outputs.	Pass
Display probability breakdown	Safe and Phishing percentages are visible in the popup.	Pass
Expose contextual characteristics	Email Characteristics are visible in the scrolled views.	Pass
Expose phishing indicators	Warning or no-indicator states are shown depending on the message.	Pass
Explain the detection method	The popup states that DeBERTa contextual/semantic analysis is supplemented by rule-based indicators.	Pass
Operate in a controlled held-out test environment	The localhost dataset interface, 50-message inbox and retained ground truth are visible.	Pass

8. Security, Privacy and Deployment Considerations

- The demonstrated backend is local, reducing unnecessary transmission of test email content to an external service during development.
- The controlled dataset page uses synthetic sender metadata and inert links, avoiding interaction with live phishing infrastructure.
- The browser extension should request only the permissions required for the intended pages and API endpoint.
- Production deployment would require stronger origin restrictions, transport security, authentication/authorisation where appropriate, logging controls and privacy review.
- Email content can contain personal or sensitive information; a production system should minimise retention and clearly define data handling.
- The current evidence validates the controlled prototype, not unrestricted access to arbitrary mailboxes.

9. Limitations and Evidence Boundaries

- The screenshots validate the extension in the controlled Gmail-style dataset page; they do not prove robust extraction across every live Gmail DOM layout or future Gmail interface change.
- The extension itself does not classify emails; it depends on the backend service and model being available.
- The displayed security indicators are supplementary and must not be described as the final model prediction.
- The screenshot set is qualitative functional evidence and should not be used to calculate aggregate accuracy, precision, recall or F1-score.
- Sender and subject values in this controlled interface are synthetic test metadata; the evidence does not constitute sender-authentication, SPF, DKIM or DMARC validation.
- False positives and false negatives remain possible even when individual screenshots show high confidence.
- Browser-extension testing should remain separated from formal model evaluation so that UI/integration success is not confused with statistical generalisation.

10. Relationship to the Other Project Artefacts

This browser-extension artefact complements the Streamlit artefact and the held-out dataset-testing artefact. The Streamlit interface demonstrates direct manual email submission and SHAP explainability, while this artefact demonstrates browser-based extraction and result presentation beside an email. Together they show that the backend detection capability can be accessed through more than one interface while preserving a consistent classification workflow.

11. Conclusion

The browser-extension evidence demonstrates a working controlled integration between a mail-client-style page and the project's phishing detection backend. Across twelve supplied tests, the

extension extracts the currently displayed email, invokes analysis, displays Safe/Phishing probabilities, reports email characteristics and contextual indicators, and explains the detection approach. The artefact therefore provides practical implementation evidence for the project's browser-integration stretch goal while maintaining an appropriate distinction between controlled prototype validation and production Gmail deployment.